

NAME: _____

PERIOD: _____

DATE: _____

Two Samples, One Difference

AP Statistics · Unit 4 · Topic 4.6 · B: 2027-01-22 · E: 2027-03-10

[STAR] TODAY'S PROBLEM

Suppose parcel pickup waits have the shown population parameters. North waits are approximately normal; South waits are strongly right-skewed. Independent SRSs are taken without replacement, and the service wants $P(\bar{X}_N - \bar{X}_S > 2.5)$.

Rina: “South is skewed, so I would not use a normal model for the difference.”

Oscar: “North is normal and the combined sample size is 60, so the normal model is valid; the probability is 0.0273.”

Locker samples

Locker	n	μ	σ
North	12	6.0	1.8
South	48	4.8	3.0

FIRST TAKE — before the video (5 min)

Identify one argument you would retain and one that needs more evidence. Cite a population shape or sample size.

[BOOK] JUSTIFY EACH SAMPLE MEAN BEFORE THEIR DIFFERENCE (keep on the page)

For independent random samples, $\mu_{\bar{X}_1 - \bar{X}_2} = \mu_1 - \mu_2$ and $\sigma_{\bar{X}_1 - \bar{X}_2} = \sqrt{\sigma_1^2/n_1 + \sigma_2^2/n_2}$ when each sample meets its 10% condition. Justify each component's shape: a normal population supports a normal sample mean even for small n , while the Central Limit Theorem supports a sample mean from a nonnormal population when its own $n \geq 30$. The two groups need not use the same pathway. Interpret probabilities as outcomes from repeated pairs of samples.

Finish after the video:

DOK 1 (a) State the mean of $\bar{X}_N - \bar{X}_S$ and translate the event above into context.

DOK 2 (b) Verify both 10 percent conditions, compute the sampling-distribution SD, and calculate the probability under a normal model.

DOK 3 (c) Adjudicate both accounts and decide whether 0.0273 is warranted. Justify each sample mean separately using its shape and sample size, coordinate independence, then use a South sample of 8 to expose the consequence of any rejected shortcut.

Frame: For North, the relevant shape evidence is _____; for South, it is _____.

Frame: If the South sample were 8, the model would be _____ because _____, changing trust in _____.

EXIT — turn this in

One thing the video changed about my first take: _____